

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

Roll No. :S040	Name:ADITYA SOPTE
Class :SY IT	Batch :
Date of Assignment :12-08-2026	Date/Time of Submission :12-08-2026

Part – A Theory

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

CODE:

```
kj.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/kj.py (3.14.6)
File Edit Format Run Options Window Help
import pandas as pd

data = {
    "name": ["Aditya", "Rugved", "Suraj"],
    "marks": [90, 80, 70],
    "grade": ["A", "A+", "B"]
}
df = pd.DataFrame(data)

newdf = df.filter(items=["name", "marks", "grade"])

print(newdf)
|
```

OUTPUT:

```
>>> |
      |
      |      name  marks  grade
      |  0  Aditya    90     A
      |  1  Rugved    80    A+
      |  2   Suraj    70     B
```

5b. Write a python code to show statistical information on given data set.

CODE:

```
kj.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/kj.py (3.14.6)
File Edit Format Run Options Window Help

import statistics as stat

print(stat.mean([1, 3, 5, 7, 9, 11]))
print(stat.median([1, 3, 5, 7, 9, 11]))
print(stat.mode([1, 3, 5, 7, 9, 11]))
print(stat.variance([1, 3, 5, 7, 9, 11]))
print(stat.pstdev([1, 3, 5, 7, 9, 11]))
|
```

OUTPUT:

```
=====
6
6.0
1
14
3.415650255319866
>>>|
```

5c. Write a python code to create pandas series from dictionaries.

CODE:

```
kj.py - C:/Users/IT-23/AppData/Local/Programs/Python/Python314/idle/kj.py (3.14.6)
File Edit Format Run Options Window Help

import pandas as pd

dictionary = {'Aditya': 20, 'Rugved': 21, 'Suraj': 19}

series = pd.Series(dictionary)

print(series)
|
```

OUTPUT:

```
Aditya    20
Rugved    21
Suraj     19
dtype: int64
>>>|
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

CODE:

```
import pandas as pd

dict = {'name':["Kaif", "Jasad", "Aryan", "Aditya"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[60, 70, 65, 70]}

df = pd.DataFrame(dict, index = [True, False, True, False])

print(df.loc[False])
```

OUTPUT:

```
>>> |
```

	name	degree	score
False	Jasad	BCA	70
False	Aditya	MBA	70

PART – C Curiosity Question: